

Foam rolling and roller massage effects and mechanisms

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Introduction

Massage has been used for millennia as a relaxation technique, to decrease musculoskeletal and myofascial pain, relieve or cure illnesses, increase range of motion (ROM) and enhance performance (Behm, 2018). Historical records of massage originate from China (the Yellow Emperor's *Classic Book of Internal Medicine* was written around 2700 BC), Egypt (Egyptian tomb paintings showing massage as a medical therapy around 2500 BC) and India (records of massage date from 1500–500 BC). The western world adopted these techniques relatively recently, with Per Ling from Sweden incorporating hand stroking into the Swedish Movement System in the early 1800s. Over the last 20 years, foam rollers and roller massagers have become popular as self-massage therapy devices (Figure 27.1) (MacDonald et al., 2013; Sullivan et al., 2013; Halperin et al., 2014; Macdonald et al., 2014; Aboodarda et al., 2015; Bradbury-Squires et al., 2015; Pearcey et al., 2015).

Foam rollers and roller massagers are often branded as “self-myofascial release” devices, reducing or helping to reduce (myo)fascial restrictions. These fascial restrictions are reported to occur in response to injury, disease, inactivity and inflammation (Schleip 2003a,b). It is commonly reported that when fascia becomes inelastic and dehydrated, it binds around the traumatized areas, resulting in fibrous adhesions (Schleip 2003a,b). These myofascial adhesions may generate “hypersensitive tender spots”, also known as trigger points. Fibrous adhesions can



Figure 27.1

Illustrations of (A) a foam roller and (B) a roller massager, with permission from Performance Health.

be painful, initiate abnormal muscle mechanics (i.e. neuromuscular hypertonicity, decreased strength, endurance and motor coordination) and decrease soft-tissue extensibility, adversely affecting ROM and muscle length (Schleip 2003a,b).

Rolling involves moving over the muscle and fascia with a dense foam roller or roller stick (roller massager), with either small excursions

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or longer movements from the proximal to the distal portion of the muscle or vice versa. Foam rollers now appear in many forms. Some rollers are made from polyvinyl chloride pipe surrounded by neoprene foam while others may be made from closed cell foam. Roller sticks or roller massagers are typically composed of dense foam wrapping around a solid plastic cylinder. There are many types, with some having a ridged design that is supposed to allow for both superficial and deep-tissue massage (Sullivan et al., 2013; Bradbury-Squires et al., 2015). Others have cells (squares or circular shapes) of different sizes and hardness, with recent modifications including vibration.

The few studies examining vibrating rollers have shown some advantages over non-vibrating rollers, such as inducing greater pain tolerance (Cheatham et al., 2017; Han et al., 2017), lower extent of proprioceptive errors (Lee et al., 2018) and some specific improvements in joint ROM (greater hip flexion and internal rotation but similar hip extension and external rotation; Han et al., 2017). Possible mechanisms underlying these effects may be related to vibration-induced increases in tissue temperature, blood flow, or neurogenic potentiation through augmented muscle spindle activity and gamma motoneuron activity. However, more research is needed to investigate this type of hybrid roller.

Rolling-induced pain inhibition

Rolling has been shown to reduce chronic pain such as muscle or myofascial tender spots (Aboodarda et al., 2015; Wilke et al., 2018), relatively short term pain associated with delayed onset muscle soreness (DOMS) (Macdonald et al., 2014; Pearcey et al., 2015) as well as acute pain induced by high frequency, electrical muscle (tetanic) stimulation (Cavanaugh et al., 2017).

DOMS was induced in two similar studies with 10 sets of 10 squat repetitions to failure. Subjects in the first study were examined 24 and 48 h after the exercise-induced muscle damage (Pearcey et al., 2015). In the second study, participants were also measured at 72 h post-squat exercise protocol (MacDonald et al., 2014). In both studies, the experimental group rolled before testing and the controls did not. Rolling decreased pain perception by 18–24% and 16–19% at 24 and 48 h post-exercise, respectively, in both studies. Other performance tests that were impaired by DOMS, such as muscle activation, vertical jump, sprint time and strength endurance, showed less impairment with rolling.

While exercise-induced muscle damage pain may persist for a week in response to muscle damage and inflammatory responses (Behm et al., 2001), muscle tender points may persist for months. Rolling has been demonstrated to also decrease muscle tender point pain. Aboodarda et al. (2015) had a massage therapist find and identify the most sensitive muscle tender point on participants' calf (triceps surae/plantar flexors). The painful calf was then either massaged by the therapist, rolled or not rolled (control). In addition, the contralateral calf was also rolled. Rolling or massaging the affected calf decreased muscle tender point pain sensitivity. Curiously, rolling the contralateral calf also reduced pain sensitivity. Hence, the description of rolling as a self-myofascial release technique cannot possibly explain the full array of mechanisms, if there was no need to even touch the painful calf.

With a similar study, the pain associated with electrically induced (tetanic muscle contractions) high frequency stimulation was also diminished without manipulating the affected limb. Cavanaugh et al. (2017) induced high levels of acute pain by stimulating the tibial nerve

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(activates plantar flexors) with maximal and sub-maximal (70%) intensity high frequency (50 Hz) electrical stimulation (tetanus). Similar to the previous study, the affected calf was either rolled or not (control), while the contralateral calf was also rolled. Each time muscle stimulation was inflicted, pain sensitivity increased by ~10%. Whether the affected or contralateral limb was rolled, the result was a decrease in pain. It seems evident that a systemic or global neural response contributed to the pain inhibition.

Pain receptors (nociceptors) are widespread in muscle, fascia and skin (Kosek et al., 1995). Even light rolling can increase the sensitivity of superficial nociceptors (Aboodarda et al., 2015). This afferent stimulation, whether applied to the affected limb or a contralateral limb, may be related to the gate control theory of pain (Melzack and Wall, 1965; Moayedid and Davis, 2013), diffuse noxious inhibitory control (Mense, 2000) and parasympathetic nervous system alterations (Weerapong et al., 2005). The gate control theory involves stimulation of lower velocity, myelinated group III and IV afferents nerve fibers (via activation of skin, fascia and muscle mechanoreceptors, metaboreceptors and proprioceptors), which transmit the signals from ascending nociceptors via small diameter A δ fibers to the periaqueductal gray nucleus (Moayedid and Davis, 2013). Pain suppression can arise peripherally from the subsequent activation of type I and II (i.e. tactile, pressure, vibration) higher velocity afferents that may impede the transmission of the slower type III and IV afferents (block the gate). Analgesia can also emanate supraspinally by activating opioid receptors, which inhibit pain with serotonergic and noradrenergic neurons (Pud et al., 2009).

A second pain inhibitory pathway could be ascribed to diffuse noxious inhibitory control (DNIC), which can work by counter-irritation

activating nociceptive stimuli through mechanical pressure from a non-local tissue. (Your toe hurts, so irritate your hand to decrease pain in the toe). With DNIC, non-local (i.e. if toe hurts activate the hand) receptor activation is transmitted to multi-receptive, wide dynamic range convergent neurons in the cortical subnucleus reticularis dorsalis, where pain transmission is suppressed monoaminergically (i.e. transmitters such as norepinephrine and serotonin). Pain sensitivity is then inhibited, not only with the affected location, but also at distant (non-local) areas (Mense, 2000; Pud et al., 2009).

Parasympathetic activation may also be activated by massage. Pain sensitivity would be modified due to alterations in cortisol, endorphins, serotonin and oxytocin (Weerapong et al., 2005). Suppressed parasympathetic reflexes could minimize the strain on the smooth muscles embedded in the soft tissue, thus inhibiting pain by decreasing myofascial tissue stress. Massage or rolling can lead to acute ischemic compression, which has been shown to result in reduced perceived pain in the upper trapezius muscles (Kostopoulos et al., 2008) and with neck and shoulder muscles (Hanten et al., 2000).

In addition to neural mechanisms, psychophysiological factors could also contribute to pain inhibition. Magnusson et al. (1996) has proposed that the acute or immediate increase in ROM following stretching is due to an increased stretch or pain tolerance. When an individual feels discomfort or pain for a prolonged period, as with rolling or stretching, persisting with that pain or discomfort results in the individual partially accommodating the painful or uncomfortable sensation. This may be labeled as a psycho-physiological effect, as the mechanisms could be predominately psychological or work in concert with the gate control and DNIC mechanisms.

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Rolling-induced increases in ROM

Rolling can increase ROM (MacDonald et al., 2013; Sullivan et al., 2013; Halperin et al., 2014; Macdonald et al., 2014; Aboodarda et al., 2015; Bradbury-Squires et al., 2015; Markovic, 2015; Pearcey et al., 2015; Skarabot et al., 2015; Vigotsky et al., 2015; Murray et al., 2016) for as long as 20 min (Junker and Stoggl, 2015; Kelly and Beardsley, 2016). Improved joint flexibility has even been demonstrated after 1–2 sets of 5–10 s of rolling (4.3%; Sullivan et al., 2013). Other studies have used an array of rolling durations such as 15, 20, 30, 60, 90 and 120 s. Typically, longer durations show greater increases in ROM than shorter durations. For example, the 10 s of rolling in the Sullivan et al. (2013) study provided a greater sit and reach score than 5 s. Other studies have reported that 60 s of rolling provided greater flexibility results than 20 s (16% vs. 10%, Bradbury-Squires et al., 2015), while 120 s exceeded 60 s with both foam rolling (9.8% vs. 30%) and roller massage (11% vs. 21.9%) (Monteiro et al., 2017a). ROM improvements are quite variable (3–30%) (Grieve et al., 2015; Skarabot et al., 2015). Rolling possesses some of the advantages of static stretching, such as improvements in flexibility, while generally not sharing the disadvantage of impaired performance after prolonged static stretching in isolation (without the inclusion of dynamic activities; Behm and Chaouachi, 2011; Behm et al., 2016). With most rolling studies, there is no adverse effect on subsequent performance such as muscle strength, power and balance (MacDonald et al., 2013; Sullivan et al., 2013; Halperin et al., 2014; Healey et al., 2014; Behara and Jacobson, 2015). An additional positive rolling effect that may or may not contribute to the increased ROM is the report that foam rolling reduces arterial stiffness and improves vascular endothelial function (Okamoto et al., 2014). Hence, there could be other health benefits that

need to be explored further. The effect on performance will be examined in more detail later (Table 27.1).

One of the major factors contributing to rolling-induced increases in ROM are thixotropic effects, which occur when fluids become less viscous or more fluid-like when agitated, sheared or stressed. Rolling adds shearing and compression pressure to the soft tissue and also creates friction. The rolling-induced shear stress and increased friction-related temperature of the soft tissues can decrease the viscosity of intracellular and extracellular fluid, providing less resistance to movement. For example, hyaluronan is a glycosaminoglycan within fascia that provides compression strength, lubrication and hydration, and facilitates fascial movement. It is a highly thixotropic material that, dependent upon its state (either high or low viscosity), can greatly affect fascia's resistance to movement or sliding (Cowman et al., 2015; Fede et al., 2018). An excellent example in daily life is when an individual attempts to pour some ketchup from a container/bottle. Typically, the ketchup will not flow quickly initially. However, if the container is jostled or shaken first, then opened and squeezed, then the ketchup flows smoothly. The stress and shearing action on the ketchup decreases its viscosity making it more fluid-like and thus easier to flow. Similarly, after rolling, the high viscosity fluid-induced resistance to movement is lowered, allowing the myofascia and musculotendinous tissues to elongate or lengthen to a greater degree.

It is also suggested that neural inhibition can contribute to rolling-induced increases in ROM. Myofascia, muscle and skin are densely innervated by sensory neurons (Schleip 2003a,b). Within the various skin layers (i.e. epidermis and dermis), there are a number of mechanoreceptors such as the Merkel receptor, Meissner

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corpuscle, Ruffini cylinder and Pacinian corpuscles. These mechanoreceptors possess a variety of different-sized receptor fields, adapt slowly or rapidly, and respond to different skin stimulation frequencies. Merkel disks (small receptor field) and Ruffini cylinders (large receptor field) are examples of slowly adapting receptor fields, which respond as long as the stimulus is present, whereas the Meissner (small receptor field) and Pacinian (large receptor field) corpuscles adapt rapidly and respond with a burst of activity at both the beginning and end of stimulation. They respond to a variety of stimulation frequencies ranging from 0.3–3 Hz (Merkel receptor), 3–40 Hz (Meissner corpuscle) and 15–400 Hz (Ruffini cylinder) to 10–500 Hz (Pacinian corpuscle). All of these mechanoreceptors respond to slow rolling or massage, with Ruffini cylinders and Pacinian corpuscles also responding to high-frequency vibrations. While the primary responsibilities are proprioception, Ruffini and Pacinian receptors can contribute to sympathetic activity inhibition and thus help to relax the muscles (Wu et al., 1999). Ruffini receptors are more sensitive to tangential forces and lateral stretch, which are common with rolling. Ruffini corpuscle stimulation can decrease sympathetic nervous system activity (Wu et al., 1999). Interstitial type III and IV receptors can also modulate sympathetic and parasympathetic activation (Mitchell and Schmidt, 1977). Interstitial receptors can be thinly myelinated or unmyelinated and originate from free nerve endings with both low and high threshold sensory capabilities. While they can be activated by pain, they can also act as mechanoreceptors responding to tension and pressure. This massage-induced increased relaxation, decrease in heart rate and blood pressure (Ruffini and Pacinian receptors) (Weerapong et al., 2005), in addition to vasodilatory effects (Mitchell and

Schmidt, 1977), contributes to greater flexibility capacity.

Massage (Goldberg et al., 1992; Behm et al., 2013) as well as roller massage (Young et al., 2018) have decreased the Hoffman (H)-reflex to muscle action potential wave (M-wave) ratio, signaling an attenuation of the afferent excitability of the motoneuron. The H-reflex is used as a tool to measure the afferent excitability of the motoneuron. However, this reflex electromyographic (EMG) signal can be influenced by the muscle membrane action potential and thus we cannot be certain whether any changes would be due to changes in the reflex arc (changes in the muscle spindle afferent reflex activity to the motoneuron), or are attributable to changes in the muscle membrane excitability. Thus, by normalizing or comparing the H-reflex to the M-wave, it permits us to evaluate whether the change is peripheral (muscle membrane) or central (afferent reflexes to motoneuron). The suppression of the spinal reflex excitability with the H-reflex when normalized to the M-wave has been attributed to decreased alpha motoneuron excitability and/or increased pre-synaptic inhibition. Massage has suppressed the H-reflex by 40–90% in these studies (Goldberg et al., 1992; Behm et al., 2013, Young et al., 2018).

Golgi tendon organs (GTO) (type Ib afferents) respond to musculotendinous tension and strong stretch, which can result in inhibition. Huang et al. (2010) showed that a single bout of a short-duration (10 or 30 s) massage at the hamstrings' musculotendinous junction improved ROM without an increase in passive muscle tension or EMG activity. Thirty seconds of musculotendinous massage provided greater flexibility than 10 s. It could be assumed that the tension applied to the tendon would activate the inhibitory responses of the GTO, leading to greater muscle relaxation. However, GTO effects persist for only 60 ms after the stressor desists (Trajano et al., 2017). Thus,

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the respective 5.8–11.3% ROM increases found in these studies (Huang et al., 2010; Trajano et al., 2017) immediately after massage could not be due to GTO inhibition.

Further evidence for the neural influence on improved ROM is the crossover or non-local effect of rolling. Similar to the aforementioned contralateral limb pain suppression (Aboodarda et al., 2015; Cavanaugh et al., 2017), ROM improvements after rolling have been reported with non-rolled limbs. Kelly and Beardsley (2016) unilaterally rolled the plantar flexors (3 x 30 s) and found increased dorsiflexion ROM (1.9–5.5%) with the contralateral ankle for up to 10 min post-rolling. Similarly, Killen et al. (2018) had participants perform 10 repetitions of 30 s of unilateral foam rolling of the hamstrings and discovered a small magnitude (effect size = 0.2) improvement in the contralateral hip flexors. In addition, they had a stretching condition, which induced an impairment in the contralateral knee flexion maximal voluntary isometric contraction (MVIC) force, but this was not as apparent following rolling. A recent study by Monteiro et al. (2018) compared 60 and 120 s of foam rolling with proprioceptive neuromuscular facilitation stretching of the quadriceps and hamstrings. The authors found significant shoulder flexion and extension ROM increases from immediate post-intervention to 10 min post-intervention for all protocols (main effect for time) after rolling the lower body. However, not all research consistently demonstrates this effect. Grabow et al. (2017) rolled the plantar surface of the foot but did not find any non-local effects on sit and reach scores or balance. Hence, the bulk of this evidence highlights the possibility of a musculotendinous myofascial rolling-induced global or systemic afferent neural inhibition effect resulting in a lower reflex-induced muscle tonus

throughout the body. It could also suggest that the moderate or greater discomfort associated with rolling any part of the body translates into a global pain or stretch tolerance effect permitting the individual to push through a greater ROM independent of the rolled body part.

Thus, ROM improvements with massage and rolling have been attributed to thixotropic factors, neural reflex inhibition, increased parasympathetic relaxation and a greater stretch or pain tolerance. However, the label or term “self-myofascial release technique” used to describe foam rollers and rollers massagers, implying that the major mechanism of foam rolling or roller massage is a physical removal of myofascial restrictions (or trigger points), is contentious. Schleip (2003a,b) proposed that the amount of force needed to remove fascial adhesions could exceed physiological limitations. Hence, the pressure of a body mass on a foam roller or the force applied on a roller massager with the arms would probably not be sufficient to break up these adhesions. More research is needed to investigate this possible mechanism.

Rolling recommendations for ROM

ROM improvements have been demonstrated with as little as 5–10 s of rolling, with 10 s providing significantly greater increases than 5 s (Sullivan et al., 2013). The majority of studies use multiple sets of 30–60 s bouts of rolling and, although the varying durations have not been not directly compared in most studies, 60 s of rolling tends to induce greater ROM improvements (MacDonald et al., 2013; Halperin et al., 2014; Macdonald et al., 2014; Bradbury-Squires et al., 2015; Pearcey et al., 2015). However, Monteiro and Neto (2016) did

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report greater ROM with 120 s of rolling versus 60 s. The intensity or discomfort of rolling does not have to be high. Grabow et al. (2018) demonstrated that whether rolling was conducted at 5, 7 or 9/10 on a visual analog pain scale, ROM improvements were similar in each case. An earlier study by Curran et al. (2008) showed that a multi-level rigid roller exerted greater pressures on the skin than a biofoam roller, and the authors suggested that the more rigid roller might be more beneficial. However, Grabow et al.'s 2018 study shows that high pressures are unnecessary to achieve greater ROM results. Thus harder, or more unyielding, and possibly more painful, rollers may not provide any greater benefits. In addition, other devices such as balls could also be substituted to provide a stimulus to the tissues (see Figure 27.2 for examples). Furthermore, following a typical warm-up (5 min of aerobic activity, static and dynamic stretching and sport-specific activities), when rolling was performed at 10 min-intervals after the warm-up, the augmented ROM was maintained for 30 min (Hodgson et al., 2017). This result is valuable for athletes who wait on a bench for 10-30 min before entering the game. Continuing to intermittently roll their muscles while sitting on the bench could maintain some of the benefits of the warm-up.

Rolling effects on performance

One of the advantages of rolling compared with prolonged static stretching is the lower incidence of reported subsequent performance impairments. Table 27.1 illustrates that, out of 17 studies, six reported no effect of rolling on subsequent performance (MacDonald et al., 2013; Halperin et al., 2014; Behara and Jacobson, 2015; Jones et al., 2015; Aune et al., 2018; Smith et al., 2018), five examining rolling

demonstrated improved performance (i.e. vertical jump measures, MVIC force, neuromuscular efficiency, shorter electromechanical delay; Macdonald et al., 2014; Peacock et al., 2014; Bradbury-Squires et al., 2015; Monteiro et al., 2017b; MacGregor et al., 2018), and three showed an attenuation of fatigue-induced performance impairments with rolling compared with control (Healey et al., 2014; Fleckenstein et al., 2017; Madoni et al., 2018). Only two studies reported rolling-induced impairments. Monteiro et al. (2017 c) applied 60, 90 and 120 s of rolling between four sets of knee extensions. Whereas 120 s of rolling decreased the number of knee extension repetitions by 14%, the 90 and 60 s of rolling also decreased repetition numbers by 8–9% (Monteiro et al., 2017 c). Cavanaugh et al. (2016) had participants perform three repetitions of 30 s of roller massage on the plantar flexors at 7/10 on a visual analog pain scale, and found that the force produced in the first 200 ms of a knee extension MVIC (F200) decreased by 9.5% and 19.1% post-test and 5 min post-test, respectively.

Summary

Whereas massage has been used for millennia, foam rollers and roller massagers have only recently become popular. Rolling can increase ROM for ~20 min, generally without subsequent performance deficits (with the exception of two studies (Cavanaugh et al., 2016, Fleckenstein et al., 2017)). Mechanisms underlying these ROM improvements may be related to thixotropic effects (decreased visco-elasticity), rolling-induced decreases in sympathetic activity and motoneuron excitability, and increased stretch/pain tolerance. Rolling has also been shown to reduce pain pressure thresholds with

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Figure 27.2

Illustrations of sample rolling exercises with foam roller and roller massager: (A) lower back, (B) hamstrings and (C) quadriceps (roller massager).

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Reference	Intervention	ROM changes	Performance changes
Aune et al., 2018	FR: 4 weeks of 3 x 60 s on gastrocnemius	↑9% acute ↑7% with training	No change in PF torque or drop jump RSI
Behara and Jacobson, 2015	FR: 1 min each for quadriceps, hamstrings, gluteals and gastrocnemius	↑15.6% hip flexion ROM	No significant change in VJ, KE and KF torque
Bradbury-Squires et al., 2015	RM: 5 x 20 s and 5 x 60 s on quadriceps	5 x 20 s: ↑10% 5 x 60 s: ↑16%	Lunge neuromuscular efficiency ↑3–7%
Cavanaugh et al., 2016	RM: 3 x 30 s on PF at 7/10 VAS	Not measured	F200 ↓9.5% and 19.1% post-test and 5 min post-test
Fleckenstein et al., 2017	FR: Before or after fatigue: 30 s on quadriceps, hamstrings, adductors, PF and iliotibial tract	Not measured	↓fatigue-induced losses in KE MVIC. Post-fatigue FR had greater ES than pre-fatigue
Halperin et al., 2014	RM: 3 x 30 s	↑4% ankle	↑8.2% MVIC
Healey et al., 2014	FR: 30 s on quadriceps, hamstrings, latissimus dorsi and rhomboids	Not measured	Fatigue impairments of VJ, MVIC and agility were less after FR vs. planking
Jones et al., 2015	FR: 30 s on quadriceps, hamstrings, gastrocnemius and gluteals	Not measured	No effects on VJ measures
MacDonald et al., 2013	FR: 2 x 60 s on quadriceps	Knee joint ROM 10.3–12.7%↑	No change in voluntary (MVIC) or evoked (twitch and tetanus) muscle properties
MacDonald et al., 2013	FR: 20 min at post-test, post-24 h and post-48 h after EIMD (10 x 10 squats)	↑11–13% quadriceps at post-48 h and 72 h ↑3% hamstrings at post-72 h	↓1–2% EMD at post-24 h and 48 h (beneficial effect) Attenuated VJ impairments ↑1% Impaired evoked contractile properties
MacGregor et al., 2018	FR: 2 min daily for 3 consecutive days	Not measured	↑MVIC force, ↓MVC, EMG and ↑NME with 50% KE MVIC
Madoni et al., 2018	FR: 3 x 30 s on hamstrings	↑2.6% hamstrings	Hamstrings PT ↓3.4–6.1% but decreases less than control
Monteiro et al., 2017 b	FR: 30, 60, 90 and 120 s on lateral thigh	Not measured	90 s of FR↑ overhead deep squat performance vs. 30, 60 or 120 s
Monteiro et al., 2017 c	FR: 60, 90 and 120 s on quadriceps between 4 sets of dynamic KE	Not measured	↓ no. of KE repetitions by 8.6–13.8%
Peacock et al., 2014	Dynamic warm-up vs. dynamic warm-up with FR: 5 strokes x 30 s on back, gluteals, quadriceps, hamstrings, calves and pectorals	No difference between dynamic warm-up and combined	↑performance with combined warm-up for power, agility, strength and speed
Smith et al., 2018	FR: 3 x 30 s on quadriceps, hamstrings and gluteals	FR: ↑9–13.8% Combination of DS/FR: ↑7.2–15.4%	Combination ↑more than FR for VJ No change in VJ with FR
Sullivan et al., 2013	RM: 1–2 sets of 5 + 10 s	↑4.3% sit and reach	No change in MVC force or EMG

Abbreviations: DS, dynamic stretch; EIMD, exercise-induced muscle damage; EMD, electromechanical delay; EMG, electromyography; F200, force produced in the first 200 ms; KE, knee extensors; KF, knee flexors; MVIC, maximal voluntary isometric contraction; NME, neuromuscular efficiency; PF, plantar flexors; RSI, reactive strength index; VAS, visual analog scale for pain; VJ, vertical jump; arrow up, increase; arrow down, decrease.

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acute and longer term discomfort (i.e. myofascial and muscle tender points and delayed onset muscle soreness). It is recommended to perform multiple sets of 30–120 s of rolling, performed below the maximum pain tolerance (50–90%).

Rolling may be combined with static stretching to further enhance ROM as well as applied at 10-min intervals after the warm-up to maintain the increased flexibility achieved with the warm-up (see Figure 27.3 for a summary of rolling effects).



Figure 27.3

Pictograph summary of rolling effects and recommendations.

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